Intelligent Block Copolymers of Acrylamide Derivatives



Rop Aron Kipyegon

Supervisor: Prof. Dr. Kéki Sándor

Department of Applied Chemistry
University of Debrecen

This dissertation is submitted for the degree of
Doctor of Philosophy

November 2026

I would like to dedicate this thesis to my loving family ...

Declaration

Hereby I, Rop Aron Kipyegon, declare that I prepared this dissertation within the Doctoral Council for Natural Sciences and Engineering, Doctoral School of Chemistry, University of Debrecen in order to obtain a PhD Degree in Natural Sciences / Engineering at Debrecen University. The results published in the dissertation are not reported in any other PhD dissertation(s).

Debrecen, 2026.11.10

.....

Signature of the Canditate

Hereby I, Kéki Sándor, confirm that candidate, Rop Aron Kipyegon has conducted his research under my supervision within the Macromolecular and Surface Chemistry,K/4, Doctoral Program of the Doctoral School of Chemistry between 2024 and 2028. The independent studies and research activity of the candidate significantly contributed to the results published in the dissertation. I also declare that the results published in the dissertation are not reported in any other dissertation(s). I recommend the approval of his dissertation.

Debrecen, 2026.11.10

.....

Signature of the Supervisor

Intelligent Block Copolymers of Acrylamide Derivatives

Dissertation submitted in fulfilment of the requirements for the Doctoral (PhD) degree in Chemistry

Written by **Rop Aron Kipyegon**

Prepared in the framework of the Doctoral School of Chemistry of the University of Debrecen (Macromolecular and Surface Chemistry, Programme K/4)

Dissertation advisor: Prof. Dr. Kéki Sándor

The official opponents of the dissertation:

Dr. . . . . .

Dr. . . . . .

The evaluation committee:

Chairperson: Dr. . . . . .

Members: Dr. . . . . .

Dr. . . . . .

Dr. . . . . .

Dr. . . . . .

The date of the dissertation defense: 2028.00.00

Acknowledgements

Thank everyone ⋯

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Nomenclature

Greek Symbols

μ Deviation

σ Mean

Acronyms / Abbreviations

CRP Controlled Radical Polymerization

DEA Diethylacrylamide

DLS Dynamic Light Scattering

GC-MS Gas Chromatography-Mass Spectrometry

LCST Lower Critical Solution Temperature

M-MARA Multi-Step Mass Remainder Analysis

MARA Mass Remainder Analysis

NAM N-acryloylmorpholine

NIPA N-isopropylacrylamide

Tcp Cloud Point Temperature

Abstract

This is where you write your abstract ...

Chapter 1

Introduction

1.1 What is loren ipsum? Title with math σ

Lorem Ipsum is simply dummy text of the printing and typesetting industry (see Section 1.3). Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum

The most famous equation in the world: $E^2 = (m_0c^2)^2 + (pc)^2$, which is known as the **energy-mass-momentum** relation as an in-line equation.

A *LaTeX class file* is a file, which holds style information for a particular LaTeX.

$$CIF: \quad F_0^j(a) = \frac{1}{2\pi\imath} \oint_\gamma \frac{F_0^j(z)}{z-a} dz \tag{1.1}$$

1.2 Chemistry training

This section will test some common chemistry typeface. En dash – and here — em dash For example *N*–isopropylacrylamide and H_2O and the relaxation in proton ^1H NMR and relaxation of Gd(III) complexes. Even it is important to check how paragraphs are rendered. How degrees symbol is printed out 17°C. The next is finding out about nomenclature. e.g. NIPA (N-isopropylacrylamide) is a common thermoresponsive monomer

The citations will show as [1–3]

Dispersity is given by

$$Đ = \frac{Mw}{Mn} \tag{1.2}$$

This is a new paragraph and hopefully it turns out well. How citations are rendered is also important as well as in-text

It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters, as opposed to using 'Content here, content here', making it look like readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for 'lorem ipsum' will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humour and the like).

1.3 Where does it come from?

Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature from 45 BC, making it over 2000 years old. Richard McClintock, a

Latin professor at Hampden-Sydney College in Virginia, looked up one of the more obscure Latin words, consectetur, from a Lorem Ipsum passage, and going through the cites of the word in classical literature, discovered the undoubtable source. Lorem Ipsum comes from sections 1.10.32 and 1.10.33 of "de Finibus Bonorum et Malorum" (The Extremes of Good and Evil) by Cicero, written in 45 BC. This book is a treatise on the theory of ethics, very popular during the Renaissance. The first line of Lorem Ipsum, "Lorem ipsum dolor sit amet..", comes from a line in section 1.10.32.

The standard chunk of Lorem Ipsum used since the 1500s is reproduced below for those interested. Sections 1.10.32 and 1.10.33 from “de Finibus Bonorum et Malorum" by Cicero are also reproduced in their exact original form, accompanied by English versions from the 1914 translation by H. Rackham

“Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum."

Section 1.10.32 of “de Finibus Bonorum et Malorum", written by Cicero in 45 BC: “Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit

qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?"

Chapter 2

Literature Review

2.1 Reasonably long section title

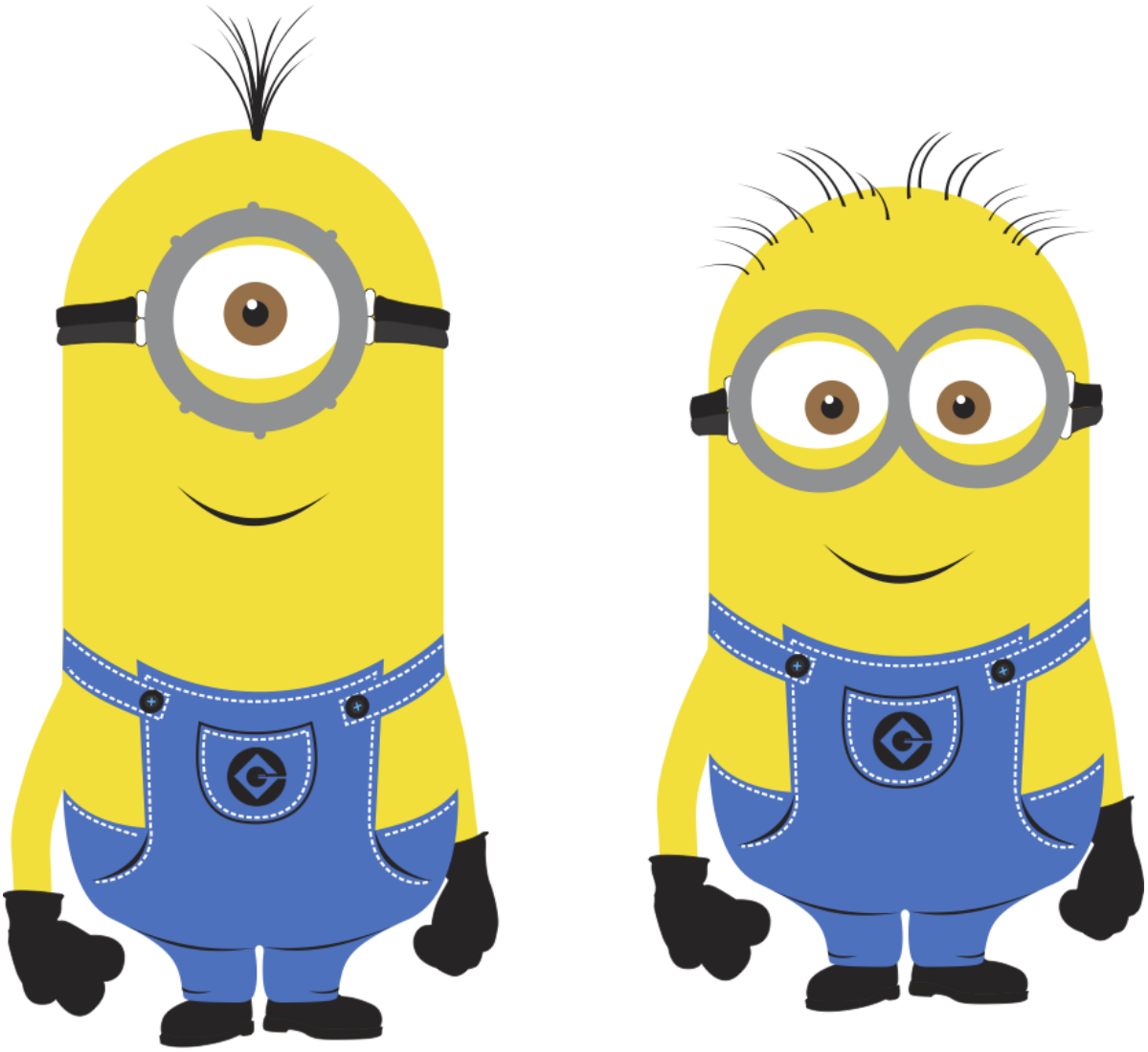


Fig. 2.1 This is just a long figure caption for the minion in Despicable Me from Pixar

Enumeration

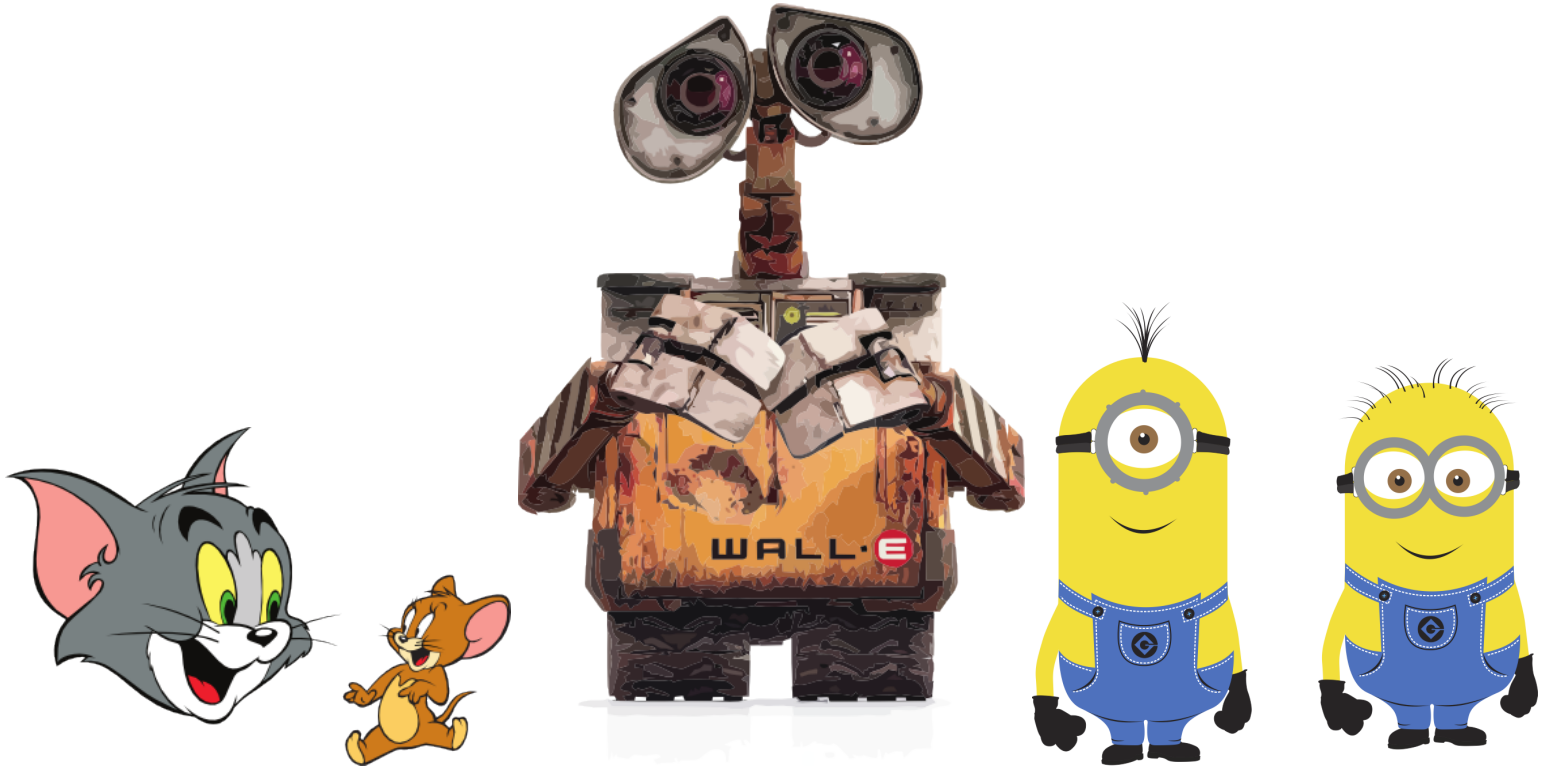
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed vitae laoreet lectus. Donec lacus quam, malesuada ut erat vel, consectetur eleifend tellus. Aliquam non feugiat lacus. Interdum et malesuada fames ac ante ipsum primis in faucibus. Quisque a dolor sit amet dui malesuada malesuada id ac metus. Phasellus posuere egestas mauris, sed porta arcu vulputate ut. Donec arcu erat, ultrices et nisl ut, ultricies facilisis urna. Quisque iaculis, lorem non maximus pretium, dui eros auctor quam, sed sodales libero felis vel orci. Aliquam neque nunc, elementum id accumsan eu, varius eu enim. Aliquam blandit ante et ligula tempor pharetra. Donec molestie porttitor commodo. Integer rutrum turpis ac erat tristique cursus. Sed venenatis urna vel tempus venenatis. Nam eu rhoncus eros, et condimentum elit. Quisque risus turpis, aliquam eget euismod id, gravida in odio. Nunc elementum nibh risus, ut faucibus mauris molestie eu. Vivamus quis nunc nec nisl vulputate fringilla. Duis tempus libero ac justo laoreet tincidunt. Fusce sagittis gravida magna, pharetra venenatis mauris semper at. Nullam eleifend felis a elementum sagittis. In vel turpis eu metus euismod tempus eget sit amet tortor. Donec eu rhoncus libero, quis iaculis lectus. Aliquam erat volutpat. Proin id ullamcorper tortor. Fusce vestibulum a enim non volutpat. Nam ut interdum nulla. Proin lacinia felis malesuada arcu aliquet fringilla. Aliquam condimentum, tellus eget maximus porttitor, quam sem luctus massa, eu fermentum arcu diam ac massa. Praesent ut quam id leo molestie rhoncus. Praesent nec odio eget turpis bibendum eleifend non sit amet mi. Curabitur placerat finibus velit, eu ultricies risus imperdiet ut. Suspendisse lorem orci, luctus porta eros a, commodo maximus nisi.

Nunc et dolor diam. Phasellus eu justo vitae diam vehicula tristique. Vestibulum vulputate cursus turpis nec commodo. Etiam elementum sit amet erat et pellentesque. In eu augue sed tortor mollis tincidunt. Mauris eros dui, sagittis vestibulum vestibulum vitae, molestie a velit. Donec non felis ut velit aliquam convallis sit amet sit amet velit. Aliquam vulputate, elit in lacinia lacinia, odio lacus consectetur quam, sit amet facilisis mi justo id magna. Curabitur

aliquet pulvinar eros. Cras metus enim, tristique ut magna a, interdum egestas nibh. Aenean lorem odio, varius a sollicitudin non, cursus a odio. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae;

Subplots

I can cite Wall-E (see Fig. 2.2b) and Minions in despicable me (Fig. 2.2c) or I can cite the whole figure as Fig. 2.2



(a) Tom and Jerry **(b)** Wall-E **(c)** Minions

Fig. 2.2 Best Animations

Chapter 3

Materials and Methods

3.1 Chemicals

3.1.1 ctable

You can use ctable package to create highly customizable tables. Check documentation for further explanation and see table 3.4

3.1.2 Another table

3.1.3 lenghty table that cover whole width

In case of double column layout, tables which do not fit in single column width should be set to full text width. For this, you need to use `\begin{table*} ... \end{table*}` instead of `\begin{table} ... \end{table}` environment. confirm if footnotes are getting printed and find a fix if not or convert to ctable

3.1.4 sideways table

You can use sideways environment to create sideways table to carry more data in a single page.

	Species I		Species II	
Dental measurement	mean	SD	mean	SD
I1MD	6.23	0.91	5.2	0.7
I1LL	7.48	0.56	8.7	0.71
I2MD	3.99	0.63	4.22	0.54
I2LL	6.81	0.02	6.66	0.01
CMD	13.47	0.09	10.55	0.05
CBL	11.88	0.05	13.11	0.04

Table 3.1 A badly formatted table

Dental measurement	Species I		Species II	
	mean	SD	mean	SD
I1MD	6.23	0.91	5.2	0.7
I1LL	7.48	0.56	8.7	0.71
I2MD	3.99	0.63	4.22	0.54
I2LL	6.81	0.02	6.66	0.01
CMD	13.47	0.09	10.55	0.05
CBL	11.88	0.05	13.11	0.04

Table 3.2 A nice looking table

Dental measurement	Species I		Species II	
	mean	SD	mean	SD
I1MD	6.23	0.91	5.2	0.7
I1LL	7.48	0.56	8.7	0.71
I2MD	3.99	0.63	4.22	0.54
I2LL	6.81	0.02	6.66	0.01
CMD	13.47	0.09	10.55	0.05
CBL	11.88	0.05	13.11	0.04

Table 3.3 Even better looking table using booktabs

	$H(Mu) + F_2$	$H(Mu) + Cl_2$
β(H)	$80.9^{\circ b}$	83.2°
β(Mu)	86.7°	87.7°

[a] for the abstraction reaction,
$Mu + HX \rightarrow MuH + X$.
[b] 1 degree $= \pi/180$ radians.
[c] this is a particularly long note, showing that footnotes are set in raggedright mode as we don't like hyphenation in table footnotes.

Table 3.4 The Skewing Angles (β) for $Mu(H) + X_2$ and $Mu(H) + HX$ [a]

Header one	Header two
Entry one[a]	Entry two
Entry three[b]	Entry four

[a] Some text; [b] Some more text.

Table 3.5 A table with notes

	Element 1[a]			Element 2[b]		
Project	Energy	σ_{calc}	σ_{expt}	Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 ± 12	780 A	1166	1239 ± 100
Element 4	500 A	961	922 ± 10	900 A	1268	1092 ± 40

Table 3.6 Example of a lengthy table which is set to full textwidth

Note: This is an example of table footnote. This is an example of table footnote this is an example of table footnote this is an example of table footnote this is an example of table footnote.

[a]Example for a first table footnote.

[b]Example for a second table footnote.

Projectile	Element 1[a]			Element[b]		
	Energy	σ_{calc}	σ_{expt}	Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 ± 12	780 A	1166	1239 ± 100
Element 4	500 A	961	922 ± 10	900 A	1268	1092 ± 40
Element 5	990 A	1168	1547 ± 12	780 A	1166	1239 ± 100
Element 6	500 A	961	922 ± 10	900 A	1268	1092 ± 40

Table 3.7 Tables which are too long to fit, should be written using the "sidewaystable" environment as shown here

Note: This is an example of table footnote this is an example of table footnote this is an example of table footnote this is an example of table footnote this is an example of table footnote.

[a] This is an example of table footnote.

[b] A very important table footnote

3.2 Equations

Equations in LaTeX can either be inline or on-a-line by itself ("display equations"). For inline equations use the `$...$` commands. E.g.: The equation $H\psi = E\psi$ is written via the command `$H \psi = E \psi$`.

For display equations (with auto generated equation numbers) one can use the equation or align environments:

$$\|\tilde{X}(k)\|^2 \leq \frac{\sum_{i=1}^{p} \left\|\tilde{Y}_i(k)\right\|^2 + \sum_{j=1}^{q} \left\|\tilde{Z}_j(k)\right\|^2}{p+q}. \tag{3.1}$$

where,

$$\begin{aligned} D_\mu &= \partial_\mu - ig\frac{\lambda^a}{2}A_\mu^a \\ F_{\mu\nu}^a &= \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f^{abc} A_\mu^b A_\nu^a \end{aligned} \tag{3.2}$$

Notice the use of `\nonumber` in the align environment at the end of each line, except the last, so as not to produce equation numbers on lines where no equation numbers are required. The `\label{}` command should only be used at the last line of an align environment where `\nonumber` is not used.

$$Y_\infty = \left(\frac{m}{\mathrm{GeV}}\right)^{-3}\left[1 + \frac{3\ln(m/\mathrm{GeV})}{15} + \frac{\ln(c_2/5)}{15}\right] \tag{3.3}$$

Chapter 4

Results and Discussion

4.1 Reasonably long section title

The SI Units for dynamic viscosity is $\mathrm{N\,s\,m^{-2}}$. The sample was prepared at a concentration of $20\mathrm{mg.mL^{-1}}$

compare

The sample was prepared at a concentration of 20mg/mL

The mass difference is 24.0575Da

4.2 Hidden section

Lorem ipsum dolor sit amet, *consectetur adipiscing elit*. In magna nisi, aliquam id blandit id, congue ac est. Fusce porta consequat leo. Proin feugiat at felis vel consectetur. Ut tempus ipsum sit amet congue posuere. Nulla varius rutrum quam. Donec sed purus luctus, faucibus velit id, ultrices sapien. Cras diam purus, tincidunt eget tristique ut, egestas quis nulla. Curabitur vel iaculis lectus. Nunc nulla urna, ultrices et eleifend in, accumsan ut erat. In ut ante leo. Aenean a lacinia nisl, sit amet ullamcorper dolor. Maecenas blandit, tortor ut scelerisque congue, velit diam volutpat metus, sed vestibulum eros justo ut nulla. Etiam nec ipsum non enim luctus porta in in massa. Cras arcu urna, malesuada ut tellus ut, pellentesque mollis risus.Morbi vel tortor imperdiet arcu auctor mattis sit amet eu nisi. Nulla gravida urna vel nisl egestas varius. Aliquam posuere ante quis malesuada dignissim. Mauris ultrices tristique eros, a dignissim nisl iaculis nec. Praesent dapibus tincidunt mauris nec tempor. Curabitur et consequat nisi. Quisque viverra egestas risus, ut sodales enim blandit at. Mauris quis odio nulla. Cras euismod turpis magna, in facilisis diam congue non. Mauris faucibus nisl a orci dictum, et tempus mi cursus.

Chapter 5

Conclusion

5.1 Let’s wind up

Summary

The PhD thesis presents ...

The dissertation consists of three major parts. In Chapter...

Work I.

Intro

In Chapter 2, brief...

Work II.

Intro

In Chapter 3, brief...

Work III.

Intro

In Chapter 4, brief...

Contributions of the thesis

In the **first thesis group**, the contributions are related to .. . Detailed discussion can be found in Chapter 2.

I/1. Contrib I.

I/2. Contrib II.

I/3. Contrib III.

I/4. Contrib IV.

In the **second thesis group**, the contributions are related to the .. . Detailed discussion can be found in Chapter 3.

II/1. Contrib I.

II/2. Contrib II.

II/3. Contrib III.

II/4. Contrib IV.

In the **third thesis group**, the contributions are related to the .. . Detailed discussion can be found in Chapter 4.

III/1. Contrib I.

III/2. Contrib II.

III/3. Contrib III.

III/4. Contrib IV.

Rop Aron K. — Publications

Contact Information

Department of Applied Chemistry
University of Debrecen
Egyetem Square 1, Hungary

aronrop18@gmail.com
www.edu.unideb.hu
06205913567

Publication Summary

Most important Publications —

1. Gergő Róth, Ákos Kuki, **Aron Kipyegon Rop**, Lajos Nagy, Zuura Kyzy Kaldybek, Máté Benedek, Miklós Zsuga, Tibor Nagy, Sándor Kéki(2025). *Rapid Copolymer Analysis of Unresolved Mass Spectra by Artificial Intelligence*,Analytical Chemistry 2025 97 (**36**), 19801-19808

Conference Proceedings

4. Róth, G. , Kuki, Á. , Nagy, T. , **Rop, AK** , Zsuga, M. , Nagy, L. , Kaldybek, KZ , Benedek, M. , Kéki, S. :*Application of algorithms and neural networks to evaluate block copolymer mass spectra* In: XXXI. International Conference on Chemistry = 31st International Conference on Chemistry, Transylvanian Hungarian Technical Scientific Society, Cluj-Napoca, 48, **2025**, (ISSN 2734-7109)
3. **Rop, AK** , Róth, G. , Pardi-Tóth, VC , Zsuga, M. , Nagy, T. , Kéki, S. : *Rapid Synthesis of Thermoresponsive PNIPA-b-PNAM Block Copolymers.* In: Young Researchers' International Conference on Chemistry and Chemical Engineering (YRICCCE V), Babeș-Bolyai University, Faculty of Chemistry and Chemical Engineering, Cluj., 50, **2025**.
2. **Rop, AK** , Nagy, T. , Róth, G. , Kuki, Á. , Ayoade-Alabi, EA , Nassozi, NAC , Zsuga, M. , Kéki, S. : *Design and Synthesis of Thermoresponsive Copolymers.* In: XXXI. Nemzetkogő Kegyészkonferencia = 31st International Conference on Chemistry, Transylvanian Hungarian Technical Scientific Society, Cluj-Napoca, 47, **2025**, (ISSN 2734-7109)
1. **Rop, AK** , Nagy, T. , Kuki, Á. , Kéki, S. : *Self-Assembly of Pluronic-Type Amphiphilic Block Copolymers.* In: 4th Young Researchers' International Conference on Chemistry and Chemical Engineering YRICCCE IV - **2023**: Program and Book of Abstracts, Hungarian Chemical Society, Debrecen., 68, 2023. ISBN: 9786156018168

References

[1] Vladimir Aseyev, Heikki Tenhu, and Françoise M. Winnik. Non-ionic Thermoresponsive Polymers in Water. In Axel H.E. Müller and Oleg Borisov, editors, *Self Organized Nanostructures of Amphiphilic Block Copolymers II*, volume 242 of *Advances in Polymer Science*, pages 29–89. Springer Berlin Heidelberg, Berlin, Heidelberg, 2010.

[2] Fang Yin, Pascale Laborie, Barbara Lonetti, Stéphane Gineste, Yannick Coppel, Nancy Lauth-de Viguerie, and Jean-Daniel Marty. Dual Thermo- and pH-Responsive Block Copolymer of Poly(*N*-isopropylacrylamide)- *block*-Poly(*N* , *N*-diethylamino Ethyl Acrylamide): Synthesis, Characterization, Phase Transition, and Self-Assembly Behavior in Aqueous Solution. *Macromolecules*, 56(10):3703–3720, May 2023.

[3] Fang Yin, Barbara Lonetti, Jean-Daniel Marty, and Nancy Lauth-de Viguerie. CO2-switchable thermoresponsiveness of poly(*N*,*N*-(diethylamino)ethyl acrylamide)-based homo- and copolymers in water. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, 674:131930, October 2023.